Typing Systems Cheat Sheet: Static vs Dynamic, Strong vs Weak

Static Typing

Type checking is done at compile time. Variables must be declared with a type.

Dynamic Typing

Type checking is done at runtime. Variables can hold values of any type.

Strong Typing

Languages enforce strict type rules. Implicit conversions are not allowed.

Weak Typing

Languages allow implicit type conversions, which can lead to unexpected behavior.

Language Comparison

Language	Typing Style	Type Safety	Behavior
Java	Static + Strong	Safe & strict	Compile-time errors for type mis
Python	Dynamic + Strong	Safe but runtime	Runtime errors for type mismatc
JavaScript	Dynamic + Weak	Flexible but risky	Implicit conversions can cause

Java Examples

```
int x = 5;
String s = x; // Error: Compile-time type mismatch
String s = String.valueOf(x); // OK: Explicit conversion
```

Python Examples

```
x = 5
s = x + "hello"  # Error: Runtime type mismatch
s = str(x) + "hello"  # OK: Explicit conversion
```

JavaScript Examples

```
let x = 5;
let s = x + "hello";  // OK: Result is "5hello"
let result = "5" - 1;  // OK: Result is 4
```

Java Pros and Cons

Pros:

- Early error detection
- Better performance
- IDE support

Cons:

- Verbose syntax
- Less flexibility

Python Pros and Cons

Pros:

- Easy to write
- Flexible
- Great for rapid development

Cons:

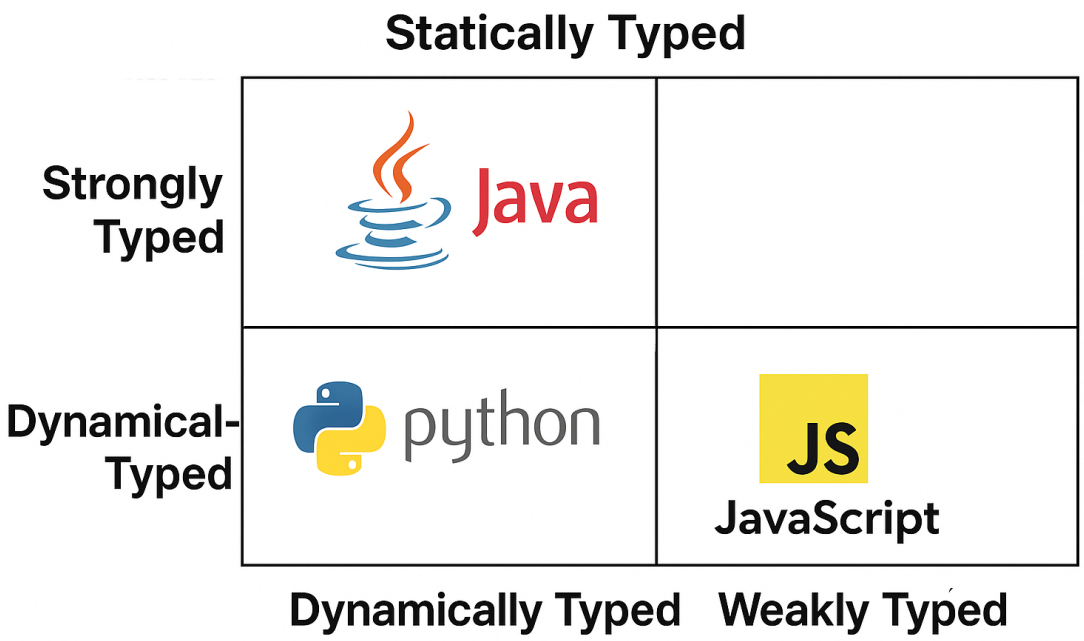
- Runtime errors
- Slower performance

JavaScript Pros and Cons

Pros:

- Flexible
- Easy to prototype
- Widely used in web

Cons:

- Type coercion bugs
- Harder to debug

Typing System Summary Diagram